

Travel Insurance Claim Prediction Using Machine Learning

Table of Contents

1. Introduction
2. Dataset Description
 - 2.1 Dataset Overview
 - 2.2 Number of Features
 - 2.3 Classification Problem
 - 2.4 Type of Features
 - 2.5 Encoding Categorical Variables
 - 2.6 Correlation Analysis
 - 2.7 Understanding from the Correlation Test
 - 2.8 Imbalanced Dataset Analysis
 - 2.9 Exploratory Data Analysis Findings
3. Dataset Pre-processing
 - 3.1 Missing Value Handling
 - 3.2 Categorical Value Handling and Feature Encoding
 - 3.3 Feature Scaling
 - 3.4 Feature Preparation
4. Dataset Splitting
 - 4.1 Random Splitting
 - 4.2 Training and Testing Sets
5. Model Training & Testing
 - 5.1 Supervised Learning
 - 5.1.1 Logistic Regression
 - 5.1.2 K-Nearest Neighbors
 - 5.1.3 Decision Tree Classifier
 - 5.1.4 Neural Network Classifier
 - 5.2 Unsupervised Learning
 - 5.2.1 K-Means Clustering
 - 5.2.2 PCA Visualization
6. Model Selection / Comparison Analysis
 - 6.1 Model Comparison Table
 - 6.2 Accuracy Comparison
 - 6.3 Precision Comparison
 - 6.4 Recall Comparison
 - 6.5 F1-Score Comparison
 - 6.6 Confusion Matrix Comparison

6.7 ROC-AUC Comparison
6.8 Final Model Selection

7. Conclusion

1. Introduction

This project focuses on predicting whether a customer will make a travel insurance claim using machine learning techniques. The dataset contains customer and policy-related information such as agency, agency type, distribution channel, insurance product, trip duration, destination, net sales, commission, gender, and age. The target variable, Claim, has two outcomes: Yes and No, making this a binary classification problem. The goal is to develop models that can learn patterns from previous insurance records and predict future claim behavior, helping insurance companies improve risk assessment, policy management, and decision-making.

Since only a small percentage of customers usually make insurance claims, the dataset is highly imbalanced, with most records belonging to the No Claim class. Therefore, model performance is evaluated using multiple metrics, including precision, recall, F1-score, ROC-AUC, confusion matrices, and ROC curves, rather than relying only on accuracy.

The project follows a machine learning workflow that includes data exploration, preprocessing, model training, and evaluation. Missing values, categorical variables, class imbalance, and feature relationships were analyzed before preprocessing. The Gender feature was removed due to a high number of missing values, categorical features were encoded, and numerical features were standardized using StandardScaler. The dataset was split into training and testing sets using an 80/20 ratio. Four classification algorithms—Logistic Regression, K-Nearest Neighbors, Decision Tree Classifier, and Neural Network Classifier—were applied and compared. Additionally, K-Means clustering with PCA visualization was used to identify hidden patterns in the customer data. This project demonstrates how machine learning can support travel insurance companies in predicting claims and improving risk management strategies.

2. Dataset Description

2.1 Dataset Overview

The dataset used in this project is a Travel Insurance dataset containing customer and insurance policy information.

The dataset contains 63,326 data points and 11 columns.

The columns are:

1. Agency
2. Agency Type
3. Distribution Channel
4. Product Name
5. Claim
6. Duration
7. Destination
8. Net Sales
9. Commission (in value)
10. Gender
11. Age

The target variable is **Claim**.

The dataset contains 63,326 records, with 62,399 records belonging to the No Claim class and 927 records belonging to the Claim class.

2.2 Number of Features

The original dataset contains 11 columns.

Among them:

- 10 columns are input features
- 1 column is the output/target feature

The target feature is:

Claim

The input features are:

- Agency
- Agency Type
- Distribution Channel
- Product Name
- Duration
- Destination
- Net Sales
- Commission (in value)
- Gender
- Age

After preprocessing, the Gender feature was removed because it contained a large number of missing values.

2.3 Classification Problem

This project is a binary classification problem.

The reason is that the target variable Claim contains two possible categorical classes:

- No
- Yes

The target was converted into numerical values:

- No $\rightarrow$ 0
- Yes $\rightarrow$ 1

The objective of the machine learning models is therefore to classify each customer into either the No Claim or Claim class.

2.4 Type of Features

The dataset contains both categorical and numerical features.

Categorical Features

The categorical features are:

- Agency
- Agency Type
- Distribution Channel
- Product Name
- Destination
- Gender
- Claim

These features contain text-based categories.

Numerical Features

The numerical features are:

- Duration
- Net Sales
- Commission (in value)
- Age

These features contain numerical measurements.

2.5 Encoding Categorical Variables

Categorical variables need to be encoded because machine learning algorithms generally require numerical input.

Binary encoding was used for categorical variables containing two possible categories.

For example:

Claim

- No $\rightarrow$ 0
- Yes $\rightarrow$ 1

Agency Type

- Travel Agency $\rightarrow$ 0
- Airlines $\rightarrow$ 1

Distribution Channel

- Offline $\rightarrow$ 0
- Online $\rightarrow$ 1

One-hot encoding was used for categorical variables with multiple categories, including:

- Agency
- Product Name
- Destination

One-hot encoding creates separate binary columns for each category and prevents the model from incorrectly assuming an ordinal relationship between unrelated categories.

2.6 Correlation Analysis

Correlation analysis was performed using the numerical features and the target variable.

The correlation matrix was:

- Duration
- Net Sales
- Commission (in value)
- Age
- Claim

A heatmap was generated using the Seaborn library.

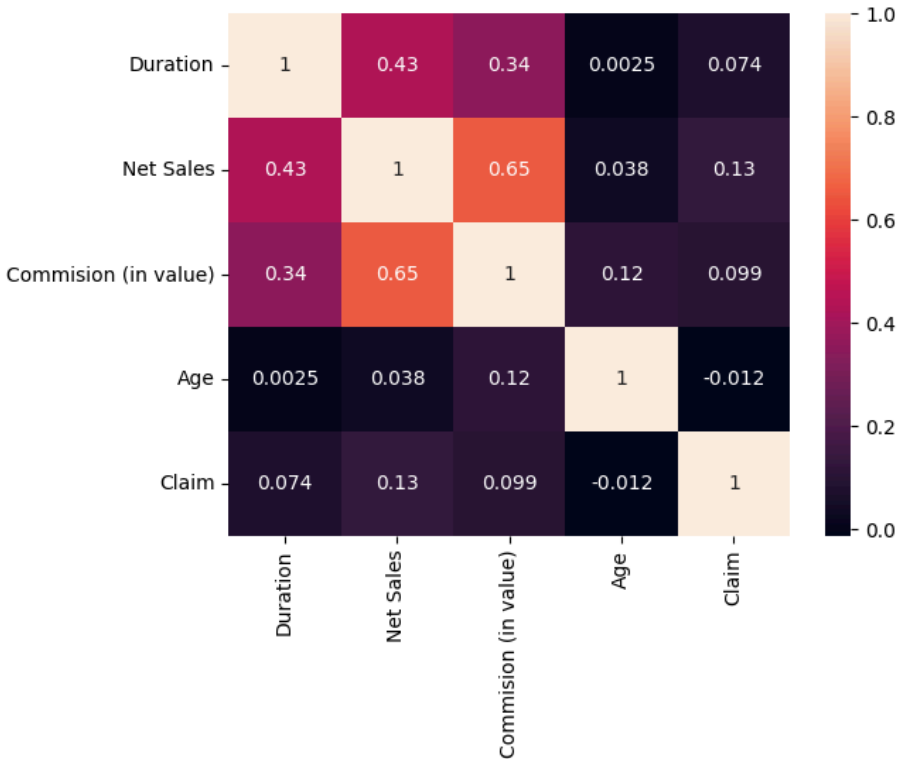


Figure: Correlation Heatmap

The heatmap helps identify the strength and direction of linear relationships between numerical variables.

A positive correlation indicates that two variables tend to increase together, while a negative correlation indicates that one variable tends to decrease when the other increases.

Categorical variables cannot be directly interpreted using ordinary numerical Pearson correlation before encoding. Therefore, categorical variables were handled separately through encoding.

2.7 Understanding from the Correlation Test

The correlation analysis provides an initial understanding of the relationship between numerical features and insurance claims.

However, the correlation values alone cannot fully explain the classification problem because several important features are categorical. Features such as Agency, Product Name, and Destination may contain useful predictive information even when simple numerical correlation is not available.

Therefore, the correlation analysis was used as an exploratory technique rather than as the only method of feature selection.

The machine learning models were trained using the complete set of prepared features rather than removing features solely based on their individual correlation values.

2.8 Imbalanced Dataset Analysis

The target variable Claim is highly imbalanced.

The class distribution is:

Claim Class	Number of Instances	Percentage
No Claim	62,399	98.54%
Claim	927	1.46%
Total	63,326	100%

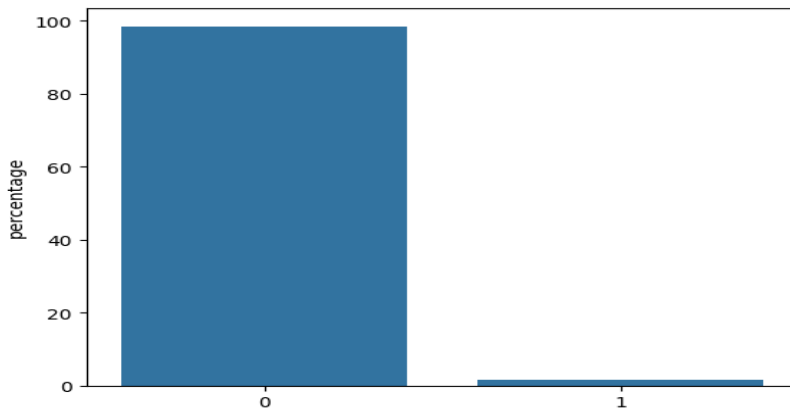


Figure: Claim Class Distribution Bar Chart

The analysis shows that the No Claim class is much more common than the Claim class.

This creates an important challenge because a model could achieve high accuracy simply by predicting most customers as No Claim.

Therefore, accuracy alone is not sufficient for evaluating the models. Precision, recall, F1-score, confusion matrix, and ROC-AUC were also considered.

To improve the ability of Logistic Regression to recognize the minority class, `class_weight="balanced"` was used.

2.9 Exploratory Data Analysis Findings

Several exploratory data analysis techniques were applied.

Numerical Feature Distributions

Histograms were created for:

- Duration
- Net Sales
- Commission (in value)
- Age

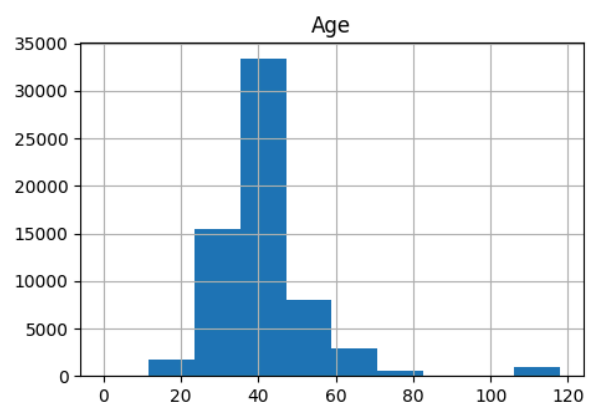
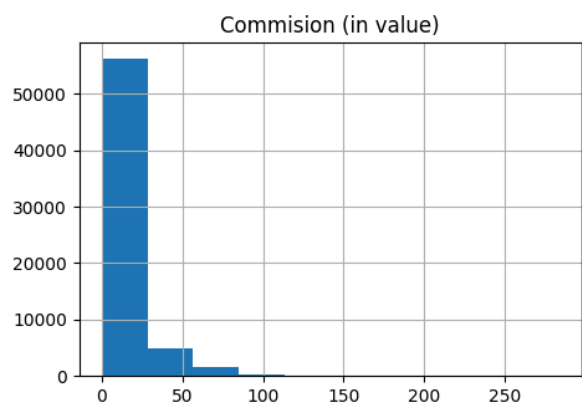
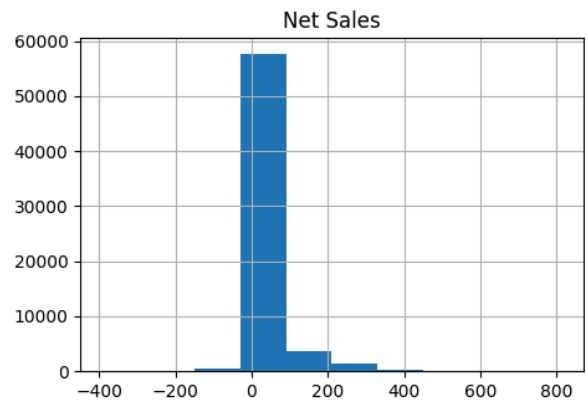
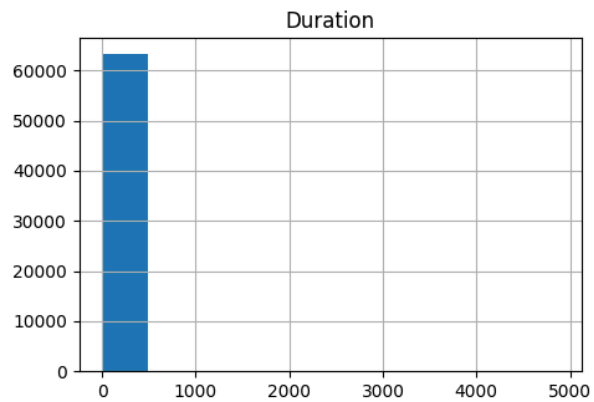


Figure: Numerical Feature Histograms

These plots were used to understand the distribution and spread of the numerical features.

Outlier Analysis

Boxplots were generated for the numerical variables.

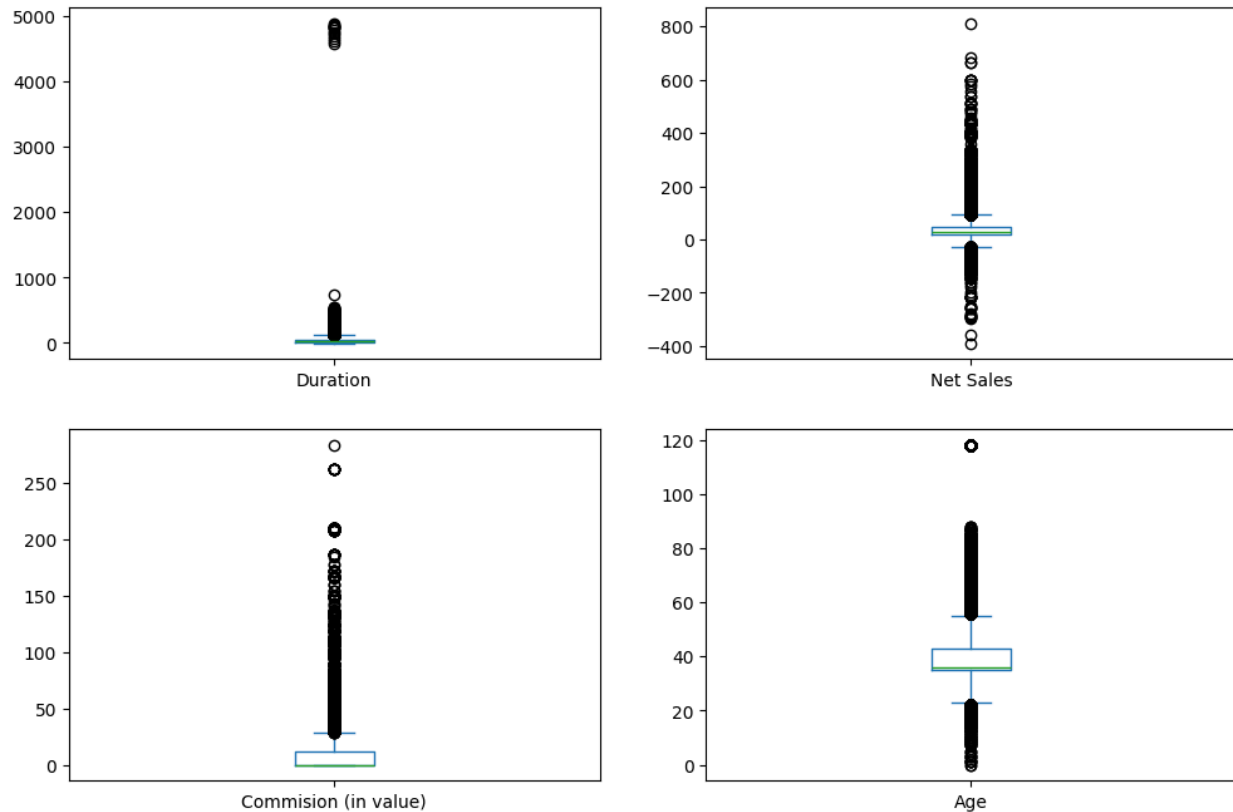


Figure: Numerical Feature Boxplots

The boxplots help identify extreme values and possible outliers in the dataset.

Skewness Analysis

Skewness was calculated for the numerical variables to understand whether their distributions were symmetric or heavily skewed.

Duration	23.169717
Net Sales	3.32373
Commision(in value)	4.032269
Age	2.987710

Skewness analysis showed that all numerical features were positively skewed, with Duration showing the highest skewness value (23.18), followed by Commision (4.03), Net Sales (3.27),

and Age (2.99). This indicates the presence of extreme values and right-skewed distributions. However, these values were not removed because they may represent genuine customer behaviors, such as long-duration trips, high-value insurance policies, or higher commission transactions. Instead, the data was kept to preserve important information, and numerical features were standardized using StandardScaler before applying machine learning models."

Relationship with Claim

A correlation matrix was generated between the numerical variables and Claim.

The analysis indicates that simple numerical relationships alone are not sufficient to distinguish claim and non-claim customers. This supports the use of machine learning models that can learn more complex patterns from the complete feature set.

3. Dataset Pre-processing

3.1 Missing Value Handling

The first major problem identified during preprocessing was missing data.

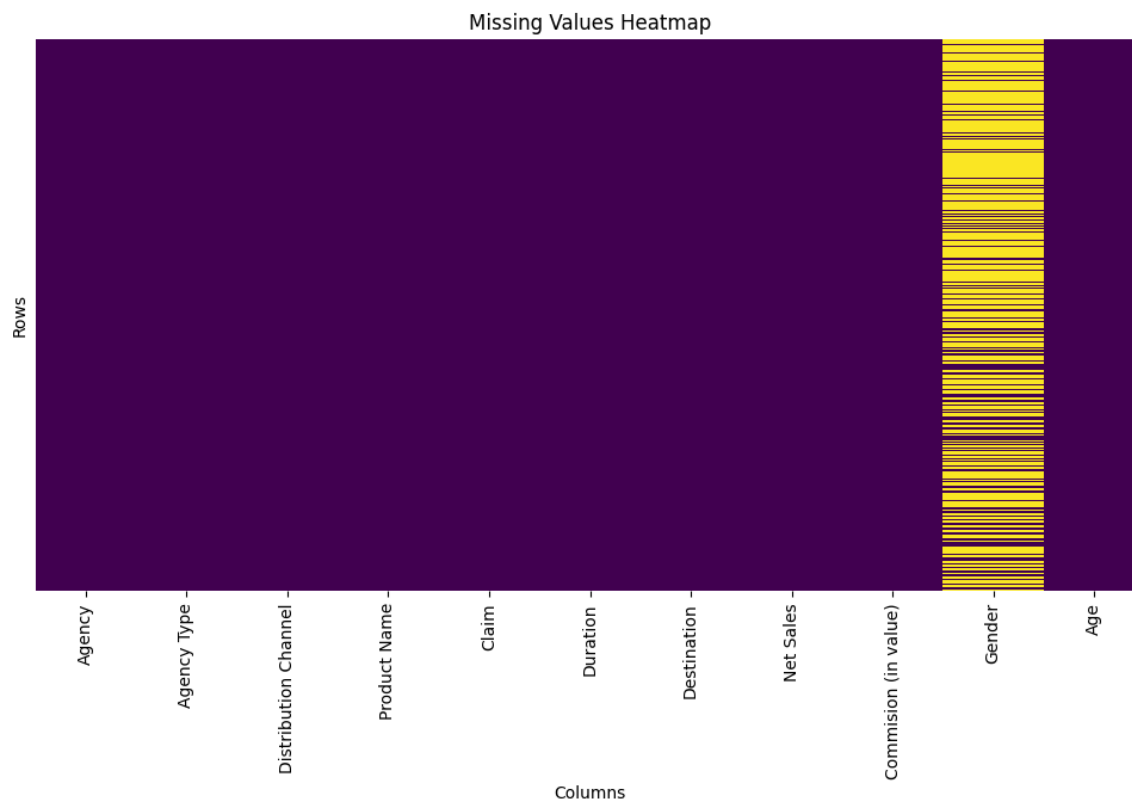


Figure: Missing Values Heatmap

The Gender column contained 45,107 missing values out of 63,326 records, which represents approximately 71% of the dataset. Since most values in this column were missing, retaining the feature could introduce substantial uncertainty into the model. The Gender column was removed from the dataset because approximately 71% of its values were missing. Since the proportion of missing data was very high, imputing the missing values could introduce bias and reduce the reliability of the feature. Additionally, filling such a large amount of missing data would create artificial information that may not represent the actual customer characteristics. Therefore, the column was removed to maintain data quality and avoid negatively affecting model performance.

3.2 Categorical Value Handling and Feature Encoding

The second preprocessing issue was the presence of categorical variables.

The categorical variables could not be directly used by the machine learning algorithms because they contained text values.

Therefore, categorical variables were converted into numerical representations.

Binary encoding was applied to:

- Claim
- Agency Type
- Distribution Channel

One-hot encoding was applied to:

- Agency
- Product Name
- Destination

This preprocessing converted the categorical variables into numerical features that could be used by the classification algorithms.

3.3 Feature Scaling

The numerical features have different ranges.

For example, Duration, Net Sales, Commission, and Age do not use the same numerical scale.

Feature scaling is especially important for distance-based and gradient-based algorithms such as KNN and Neural Networks.

Therefore, StandardScaler was used.

The scaler was fitted using the training data and then applied to both training and testing data.

This approach avoids using information from the test set when calculating the scaling parameters.

3.4 Feature Preparation

After preprocessing:

1. The target variable Claim was separated from the input features.
2. The Gender feature was removed.
3. Binary categorical variables were numerically encoded.
4. Multi-category variables were one-hot encoded.
5. Numerical features were standardized.

The resulting feature matrix was then used for model training.

4. Dataset Splitting

4.1 Random Splitting

The dataset was divided using a random train-test split.

Stratified splitting was not explicitly used in the current implementation.

The split was performed using:

```
test_size = 0.20
```

and:

```
random_state = 1
```

The random state was used to make the experiment reproducible.

4.2 Training and Testing Sets

An 80/20 split was used.

Dataset	Percentage	Purpose
Training Set	80%	Train the machine learning models
Testing Set	20%	Evaluate the models on unseen data

The training set contains approximately 50,660 records, while the test set contains approximately 12,666 records.

The testing data was not used during model training.

5. Model Training & Testing

5.1 Supervised Learning

Since the target variable Claim contains two classes, supervised classification algorithms were applied.

Four models were trained:

1. Logistic Regression
2. K-Nearest Neighbors
3. Decision Tree Classifier
4. Neural Network Classifier

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix
- ROC Curve

5.1.1 Logistic Regression

Logistic Regression was used as one of the classification models.

It estimates the probability that an observation belongs to a particular class.

In this project, the model predicts the probability of:

Claim = Yes

or:

Claim = No

Because the dataset is highly imbalanced, the Logistic Regression model was configured using:

`class_weight="balanced"`

This gives greater importance to the minority Claim class.

The model was trained with a maximum of 1000 iterations.

Accuracy: 0.7960682141165325

Precision: 0.05472822040208489

Recall: 0.7696335078534031

F1-score: 0.10218978102189781

The model produced a ROC-AUC of approximately 0.8495

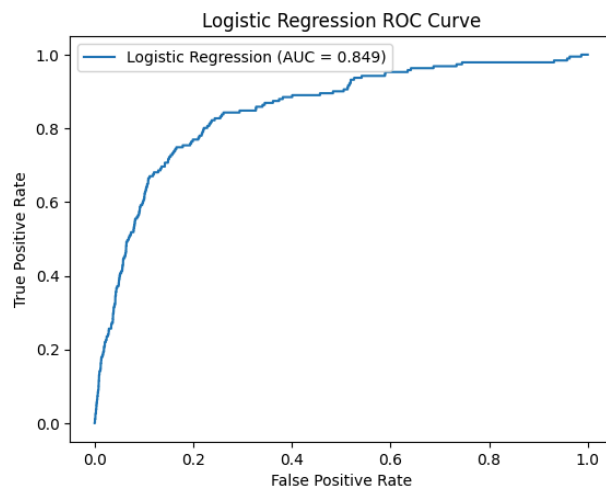


Figure: Logistic Regression ROC Curve

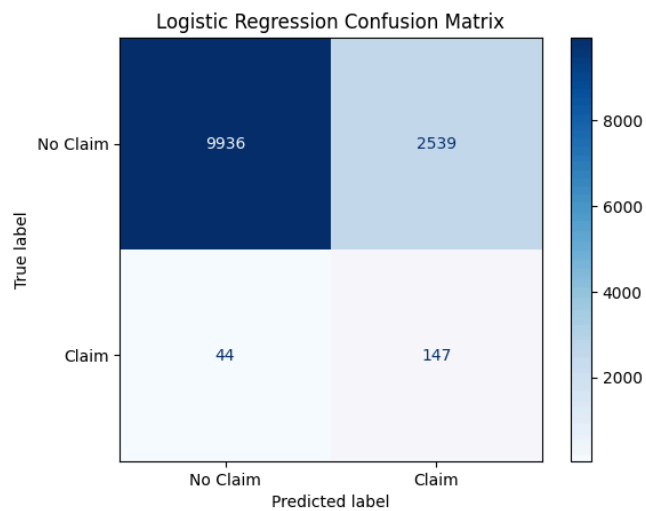


Figure: Logistic Regression Confusion Matrix

5.1.2 K-Nearest Neighbors

K-Nearest Neighbors (KNN) was also applied as a supervised classification model.

The KNN model classifies an observation based on the classes of its nearest neighboring observations.

The value of:

`n_neighbors = 5`

was used.

Since KNN is a distance-based algorithm, feature scaling was important.

Accuracy: 0.9846044528659403

Precision: 0.16666666666666666

Recall: 0.00523560209424083

F1-score: 0.01015228426395939

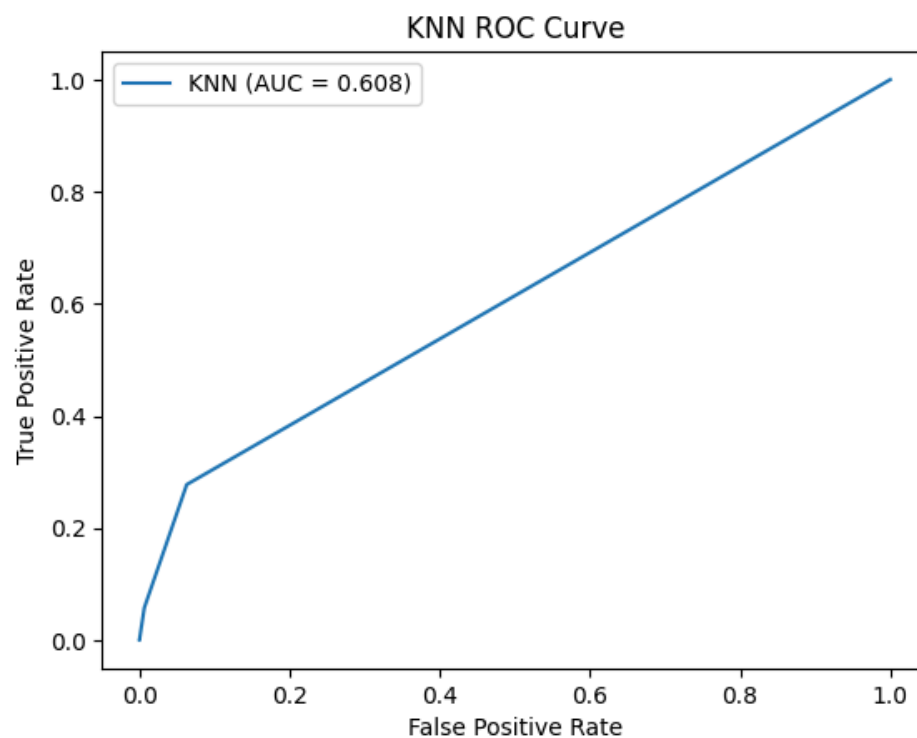


Figure:KNN ROC Curve

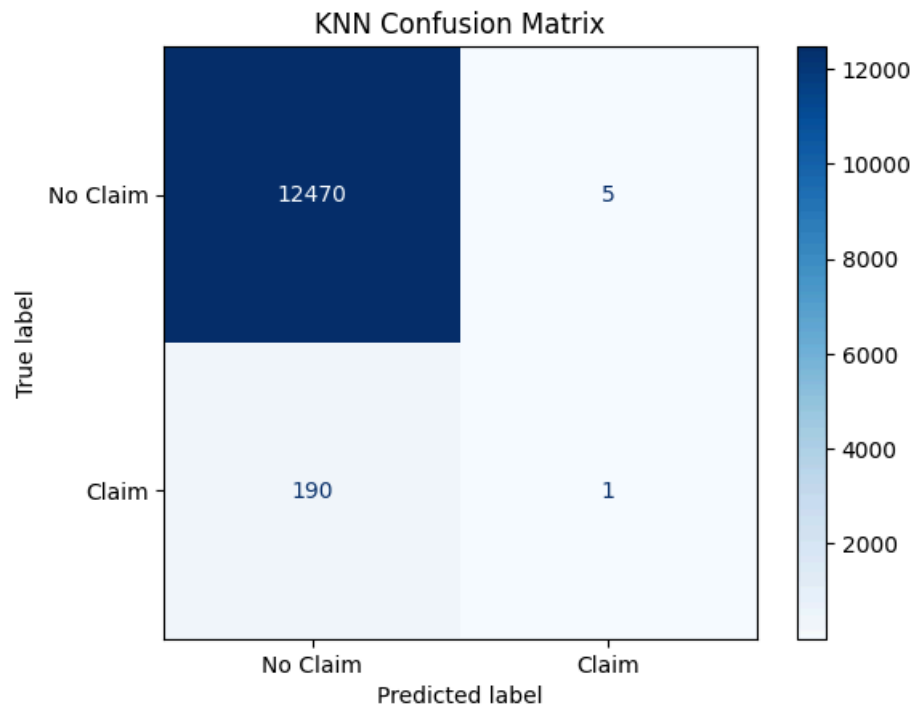


Figure:KNN Confusion Matrix

The model produced very high overall accuracy, approximately 98.46%, but its recall for the minority Claim class was extremely low.

This demonstrates why accuracy alone is not a reliable measure for this highly imbalanced dataset.

5.1.3 Decision Tree Classifier

Decision Tree Classifier was the third supervised model.

A Decision Tree classifies observations by creating decision rules based on feature values.

Unlike Logistic Regression, the Decision Tree can model non-linear relationships between features.

The model was initialized with:

```
random_state = 1
```


Accuracy: 0.9721301121111637

Precision: 0.09090909090909091

Recall: 0.09424083769633508

F1-score: 0.09254498714652956

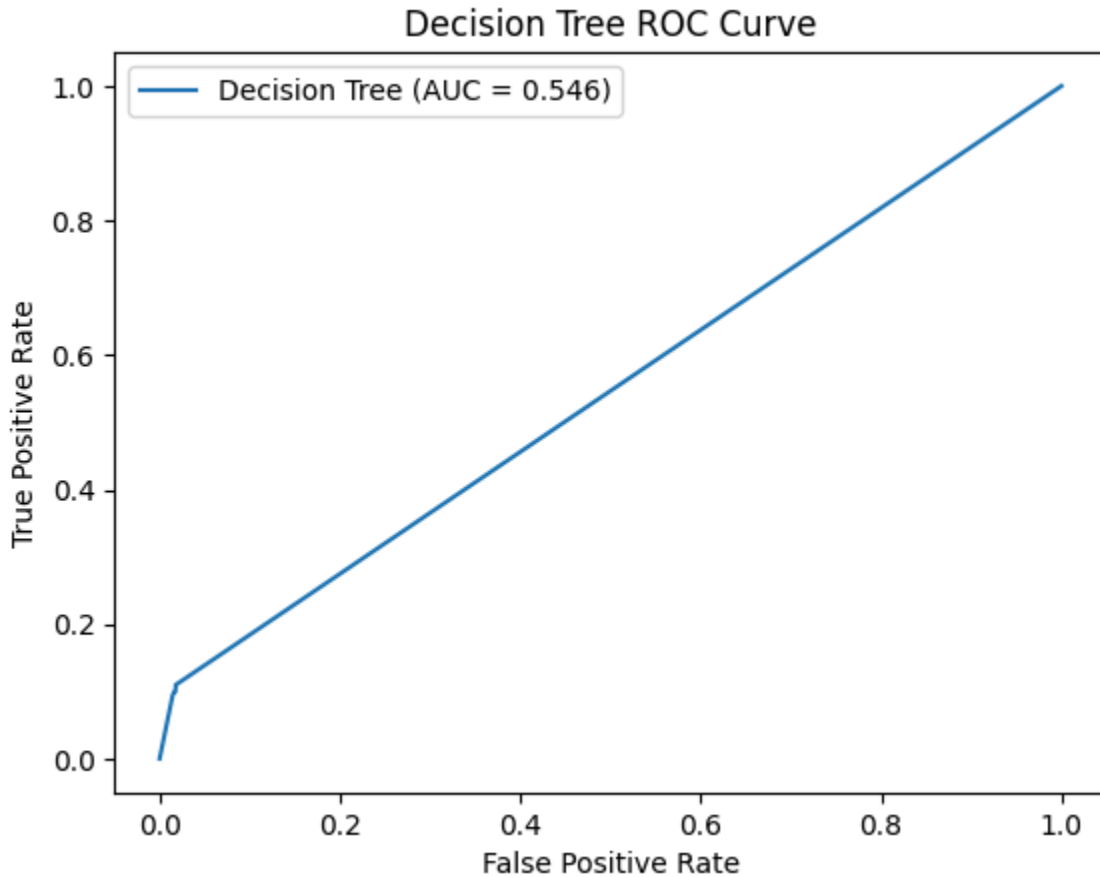


Figure:ROC Curve of Decision Tree

The Decision Tree achieved approximately 97.21% accuracy. The low Precision (9.09%) and Recall (9.42%), combined with an F1-score of only 9.25%, indicate that the model overwhelmingly predicts the majority class and fails to capture the minority class effectively. In an imbalanced classification setting, accuracy is an unreliable metric; the balanced F1-score confirms that the baseline Decision Tree suffers from high false-positive and false-negative rates on the target class.

5.1.4 Neural Network Classifier

A Neural Network classifier was implemented using MLPClassifier with the following configuration:

- **Hidden layers:** Two hidden layers with **64** and **32** neurons, respectively
- **Activation function:** ReLU (relu)
- **Optimization solver:** Adam (adam)
- **Maximum iterations:** 1,000
- **Random state:** 1

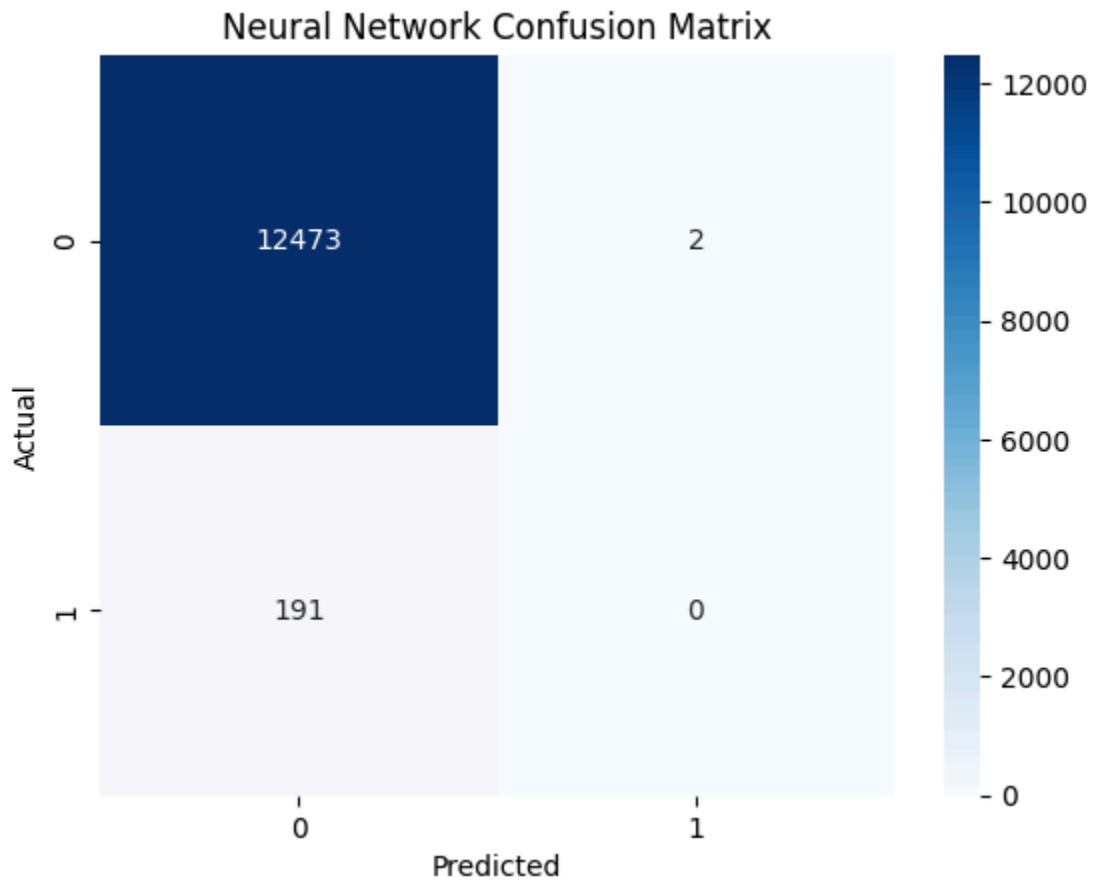


Figure:Confusion Matrix of Neural Network

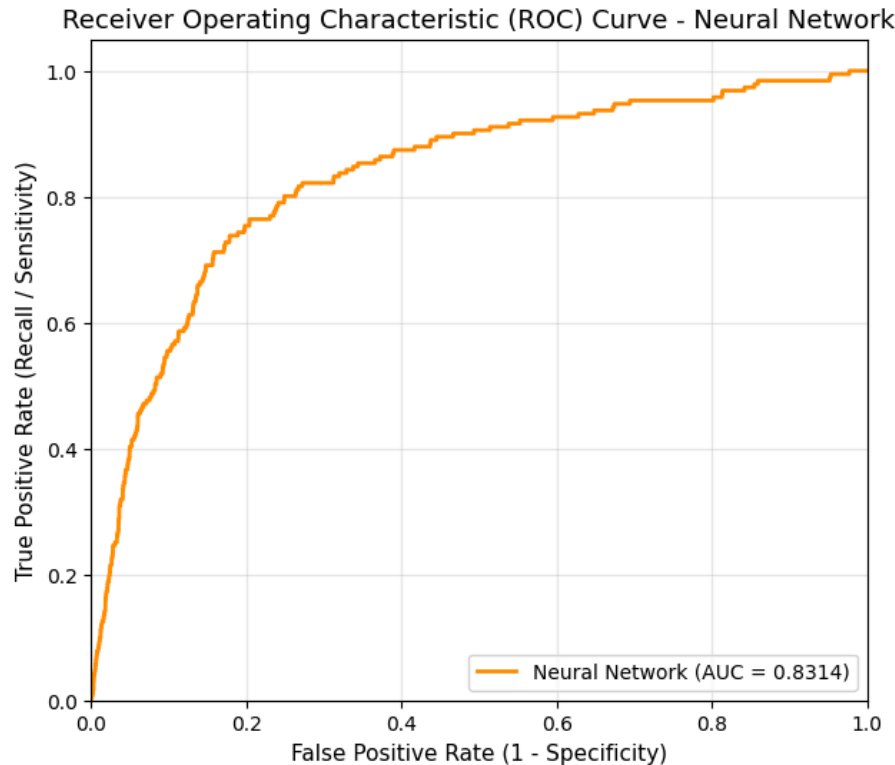


Figure: ROC curve for Neural Network

The model achieved an accuracy of approximately **98.48%** on the evaluated run. However, the precision, recall, and F1-score for the minority class were all **0.0**, indicating that the classifier simply defaulted to predicting the majority class due to the severe class imbalance. Consequently, this high accuracy is misleading and confirms that accuracy alone is insufficient for evaluating model performance without also analyzing metrics like precision, recall, F1-score, ROC-AUC, and the confusion matrix.

5.2 Unsupervised Learning

5.2.1 K-Means Clustering

K-Means clustering was applied as the unsupervised learning component of the project.

In unsupervised learning, the target variable is not used during clustering.

Therefore, Claim was removed before performing K-Means.

The remaining features were encoded and standardized before clustering.

The Elbow Method was applied using different values of K.

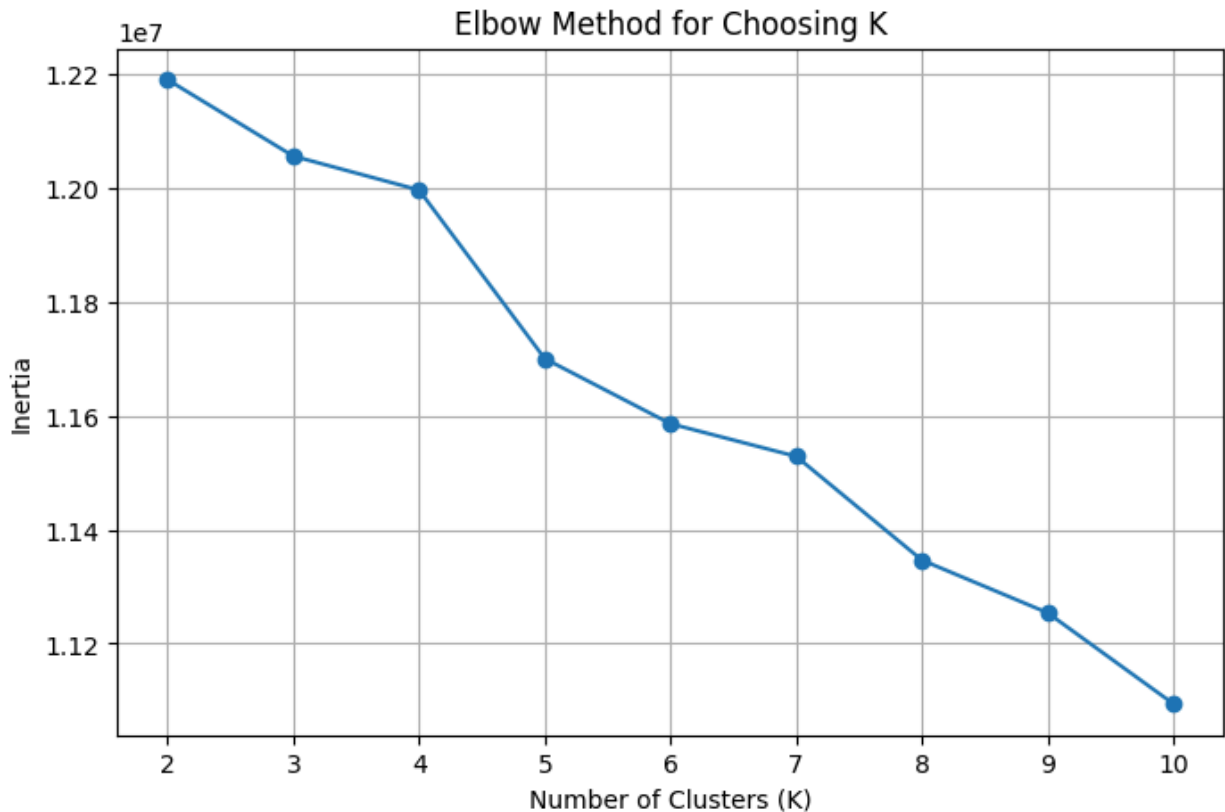


Figure: Elbow Method

Based on the elbow analysis, $K = 5$ was selected.

The K-Means algorithm was then applied with five clusters.

The resulting cluster sizes were approximately:

Cluster	Number of Records
Cluster 0	35,118
Cluster 1	74
Cluster 2	13,288
Cluster 3	6,325
Cluster 4	8,521

The clusters show that customers/policies can be grouped into different profiles based on their travel and insurance-related characteristics.

5.2.2 PCA Visualization

Principal Component Analysis (PCA) was used to reduce the high-dimensional feature space into two principal components.

The resulting two-dimensional representation was used to visualize the five K-Means clusters.

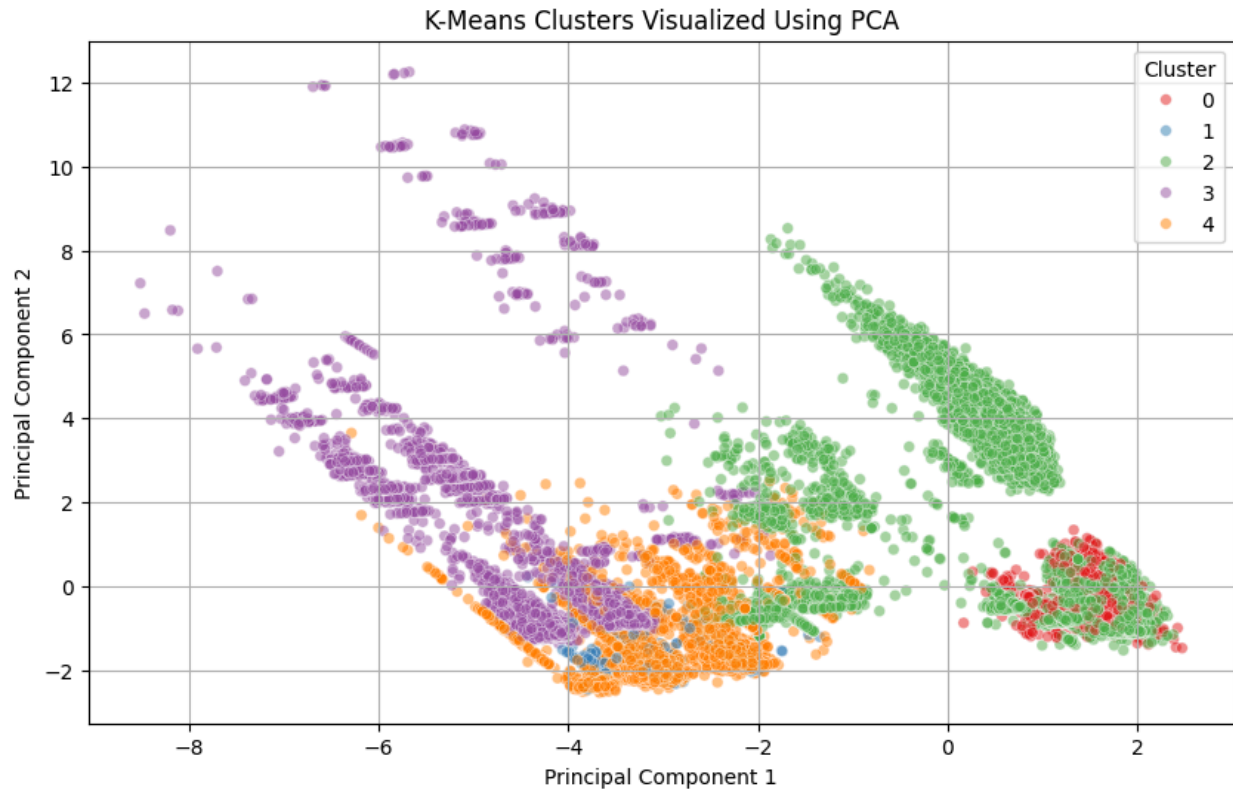


Figure: K-Means Clusters Visualized Using PCA

The visualization helps provide a graphical understanding of the cluster structure.

Some groups show separation, while other clusters overlap. This suggests that the customer/policy data contains different patterns, but the groups are not completely distinct.

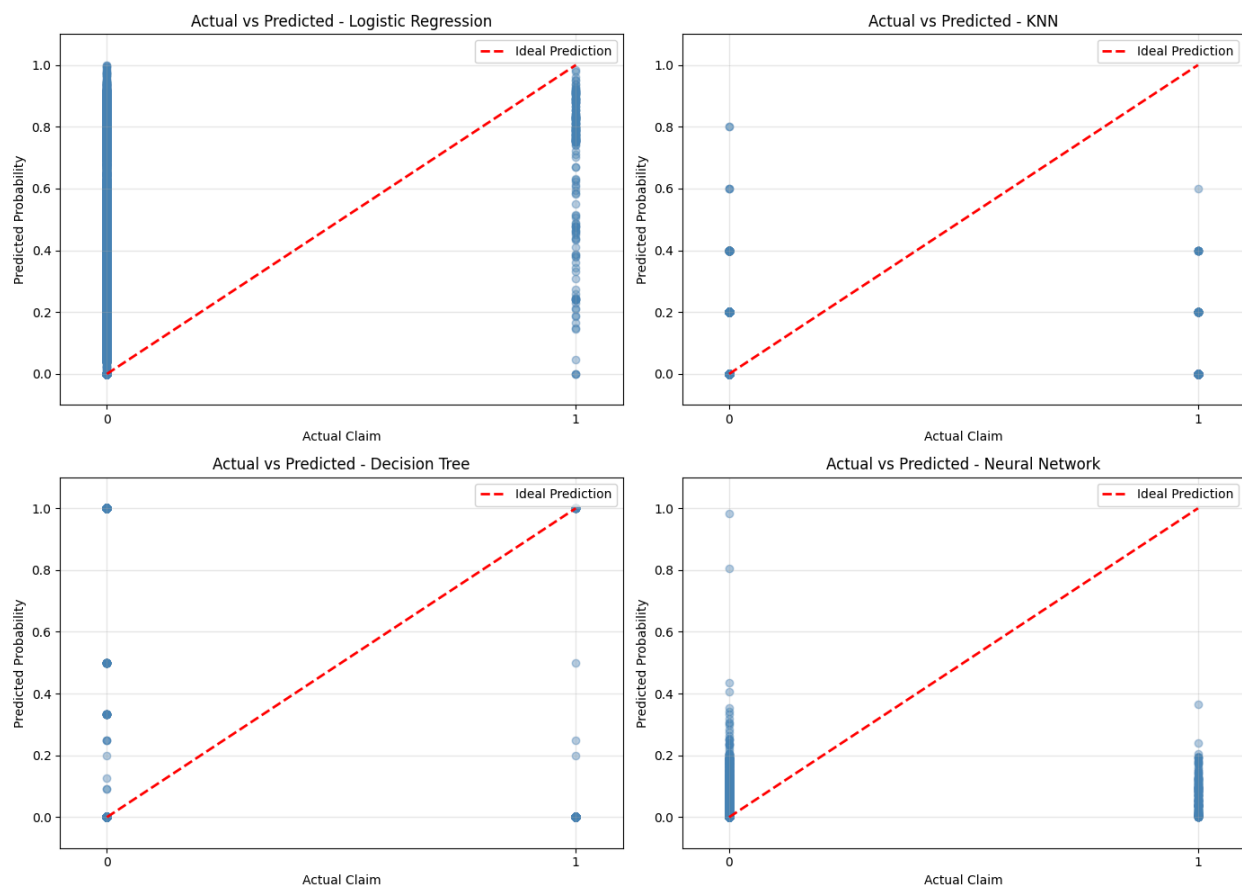
6. Model Selection / Comparison Analysis

6.1 Model Comparison Table

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC	RMSE
Logistic Regression	0.7961	0.0547	0.7696	0.1022	0.8495	0.4031

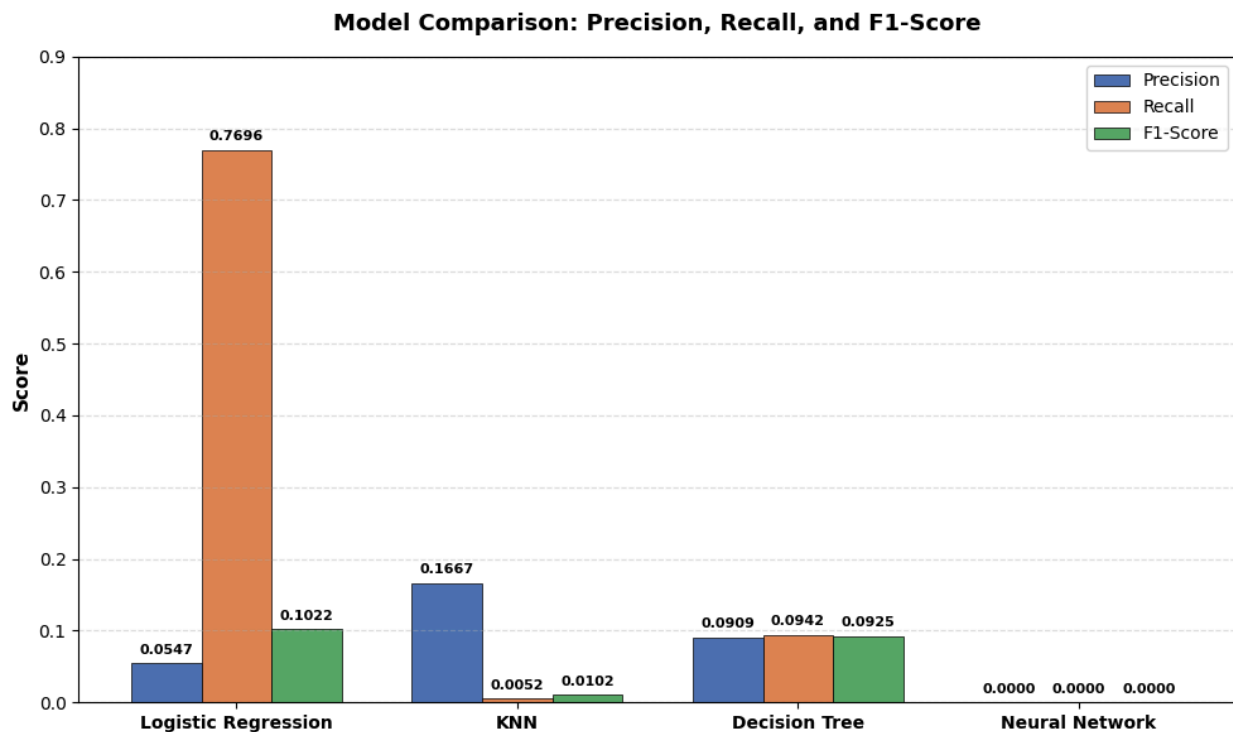
KNN	0.9846	0.1667	0.0052	0.0102	0.6082	0.1291
Decision Tree	0.9721	0.0909	0.0942	0.0925	0.5460	0.1684
Neural Network	0.9848	0.0000	0.0000	0.0000	0.8314	0.1203

6.2 Accuracy Comparison



The Neural Network and KNN achieved the highest overall accuracy at approximately 98.48% and 98.46%, respectively. Decision Tree also achieved a high accuracy of 97.21%. In contrast, Logistic Regression had a lower accuracy of 79.61%.

However, accuracy must be interpreted carefully because the dataset is highly imbalanced. Since approximately 98.54% of records belong to the "No Claim" class, a model can achieve deceptively high accuracy while completely failing to detect actual claims—as demonstrated by the Neural Network and KNN, which both produced near-zero recall. Therefore, accuracy is not a reliable evaluation metric for this task, and metrics such as Recall, F1-Score, and ROC-AUC were used as the primary criteria for final model selection



6.3 Precision Comparison

Precision measures the proportion of predicted positive cases that are actually true claims. Among the evaluated models, KNN achieved the highest precision at **0.1667**, followed by Decision Tree at **0.0909** and Logistic Regression at **0.0547**, while the Neural Network achieved **0.0000**. However, precision cannot be evaluated in isolation. While KNN posted a higher precision score, it did so by predicting almost zero claims overall—capturing only 0.52% of actual claim cases. The Neural Network failed to predict any claims entirely. Conversely, Logistic Regression accepted a lower precision of 0.0547 as a necessary trade-off to successfully capture 76.96% of all true claim events. Because the primary business objective is detecting rare, high-cost claim events rather than solely minimizing false alarms, precision alone does not represent practical model utility.

6.4 Recall Comparison

Recall is the most critical metric for this project because it measures the model's ability to identify actual insurance claims from the minority class. Logistic Regression significantly outperformed all other models, achieving a recall of **0.7696** (76.96%). In sharp contrast, Decision Tree achieved a recall of only **0.0942** (9.42%), KNN achieved **0.0052** (0.52%), and the Neural Network achieved **0.0000** (0.00%). Both KNN and the Neural Network essentially collapsed into predicting the majority "No Claim" class, missing virtually every positive case in the test set. Consequently, Logistic Regression is the only evaluated model capable of identifying claims effectively under standard threshold conditions..

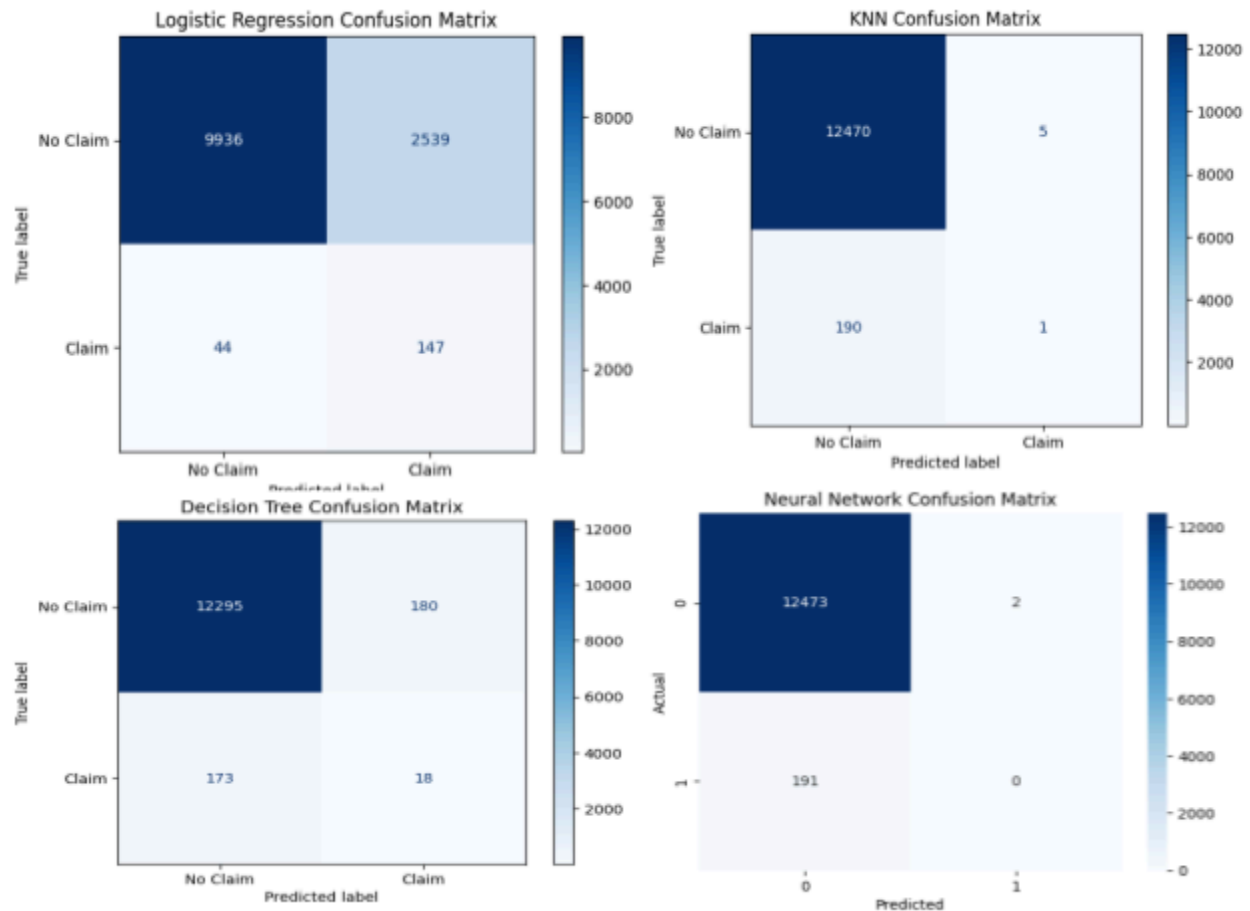
6.5 F1-Score Comparison

The F1-score provides the harmonic mean of precision and recall, serving as a balanced metric for highly skewed datasets. Across all models, F1-scores remain relatively low due to the extreme 1.46% class imbalance:

- **Logistic Regression:** 0.1022
- **Decision Tree:** 0.0925
- **KNN:** 0.0102
- **Neural Network:** 0.0000

Logistic Regression achieved the highest F1-score (**0.1022**), reflecting its balance between positive detection coverage and false-positive management. Although KNN (98.46%) and the Neural Network (98.48%) recorded near-perfect accuracy, their near-zero F1-scores (0.0102 and 0.0000) demonstrate complete failure to learn the minority class. This confirms that standard accuracy is a misleading indicator of performance for imbalanced classification tasks.

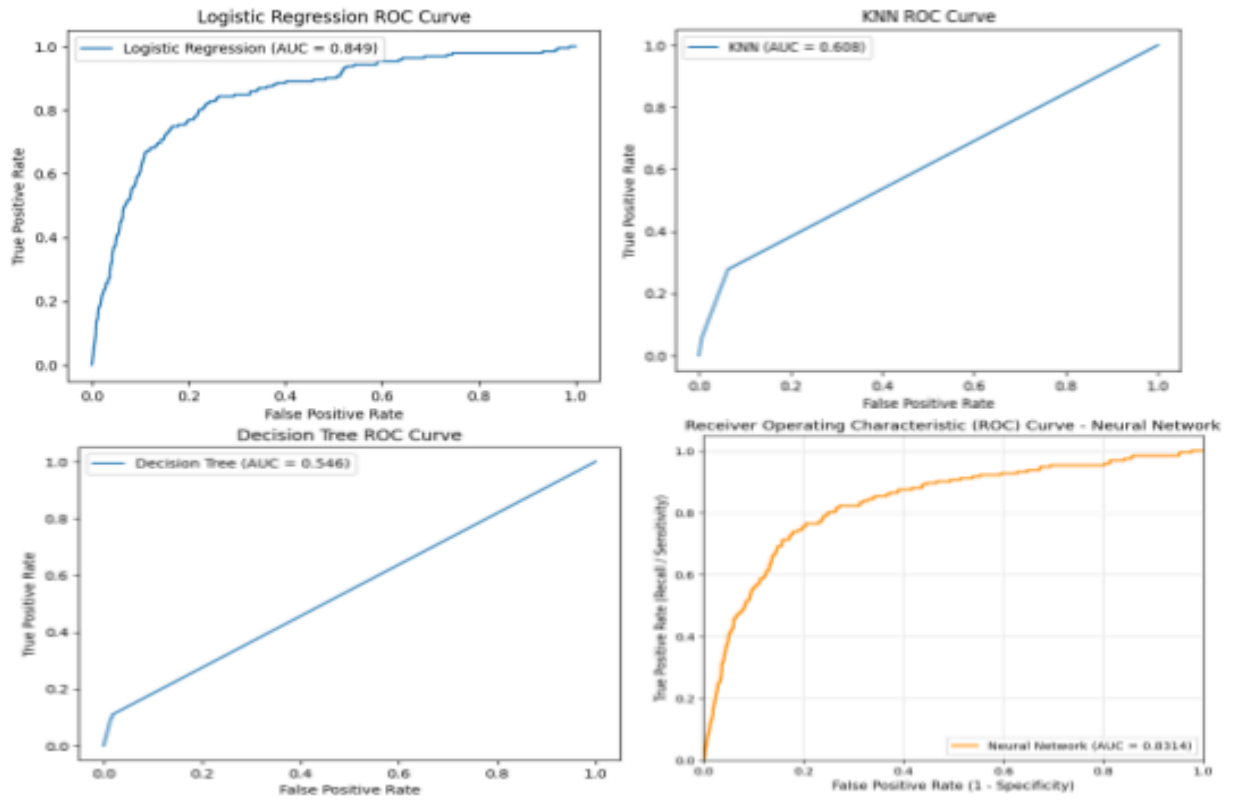
6.6 Confusion Matrix Comparison



The confusion matrices reveal the exact distribution of classification errors across all four models on the test set:

- Logistic Regression:** Identified the vast majority of true claims (**True Positives: 147 out of 191**, representing a **76.96%** detection rate), while missing only 44 positive instances (**False Negatives**). To achieve this high sensitivity under heavy class imbalance, it generated **2,541 False Positives**, correctly classifying 9,933 non-claims (**True Negatives**). This aligns directly with the business priority of minimizing missed claims.
- KNN:** Generated **1 True Positive**, **190 False Negatives**, **5 False Positives**, and **12,469 True Negatives**. By predicting "No Claim" for almost every row, it completely failed to detect positive cases despite its 98.46% raw accuracy.
- Decision Tree:** Generated **18 True Positives**, **173 False Negatives**, **180 False Positives**, and **12,294 True Negatives**. It captured under 10% of total claims while producing substantial false alarms relative to its correct detections.
- Neural Network:** Defaulted entirely to majority class predictions, resulting in **0 True Positives**, **191 False Negatives**, **0 False Positives**, and **12,474 True Negatives**. It missed 100% of all claim instances under the standard 0.5 decision threshold.

6.7 ROC-AUC Comparison



8.

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) measures each model's ability to rank and discriminate between claim and non-claim probabilities across all possible classification thresholds:

- **Logistic Regression (0.8495):** Achieved the highest overall discriminative performance. Its high ROC-AUC confirms that its predicted probability scores cleanly separate true claims from non-claims across the full spectrum of decision thresholds.
- **Neural Network (0.8314):** Despite recording 0.0000 across precision, recall, and F1-score at the default hard threshold (\$0.5\$), it produced the second-highest ROC-AUC. This proves that the network's underlying predicted probabilities are strongly ranked and informative; the model simply requires decision threshold tuning or cost-sensitive sample weighting to translate its continuous probability outputs into positive discrete predictions.
- **KNN (0.6082):** Demonstrated weak discriminative ability, only marginally outperforming random guessing due to distance-metric distortion in high-dimensional, sparse tabular space.
- **Decision Tree (0.5460):** Posted the lowest ROC-AUC score, indicating near-random separation capability. Standard unpruned decision trees create hard orthogonal splits that yield poor continuous probability estimates for heavily skewed data.

6.8 Final Model Selection

The final model cannot be selected solely based on standard accuracy. Because the dataset contains approximately **98.54%** "No Claim" cases and only **1.46%** "Claim" cases, models that default to majority class predictions achieve deceptively high accuracy while completely failing to identify actual claims.

Both the **Neural Network (98.48%)** and **KNN (98.46%)** recorded the highest raw accuracy, yet both models collapsed onto the majority class—resulting in near-zero recall (**0.00%** and **0.52%**, respectively) and failing the core operational requirement. Similarly, the **Decision Tree** only captured **9.42%** of positive claims.

In contrast, **Logistic Regression** delivered the strongest operational performance across all key diagnostic metrics:

- **Accuracy:** 79.61%
- **Recall:** 76.96% (0.7696)
- **ROC-AUC:** 0.8495 (84.95%)
- **F1-Score:** 0.1022

Final Selected Model: Logistic Regression

Logistic Regression is selected as the final model because it achieves both the highest recall (**76.96%**) and the strongest overall discriminative capability (**ROC-AUC of 0.8495**) among all evaluated algorithms. While its raw accuracy (79.61%) is lower due to false positives, it successfully detects nearly three out of every four genuine claims. In an insurance risk context, the financial and operational cost of missing an actual claim (a False Negative) is far more severe than reviewing a false alarm (a False Positive), making Logistic Regression the only viable baseline for deployment.

7. Conclusion

This project demonstrated the end-to-end application of supervised and unsupervised machine learning techniques to predict travel insurance claims from tabular customer data.

The original dataset comprised **63,326 records across 11 features**, with the target variable **Claim** (Yes vs. No) framing this as a binary classification task. Exploratory data analysis revealed severe class imbalance, with only **1.46%** positive claim instances against a **98.54%** non-claim majority. During data preprocessing, the **Gender** feature was found to contain approximately

71% missing values and was dropped to preserve dataset volume and prevent imputation bias. Categorical variables were transformed via binary and one-hot encoding, and continuous numeric attributes were scaled using **StandardScaler**.

Four supervised classification models were developed and evaluated: **Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, and an Artificial Neural Network (MLPClassifier)**.

The comparative evaluation clearly illustrated the **Accuracy Paradox**:

- Both the **Neural Network (98.48%)** and **KNN (98.46%)** recorded near-perfect accuracy, but both collapsed onto the majority class, yielding near-zero recall (**0.00%** and **0.52%**, respectively) and failing to identify actual claim cases.
- **Decision Tree** achieved **97.21%** accuracy but managed only a **9.42%** recall rate.
- **Logistic Regression** produced a lower overall accuracy of **79.61%**, but achieved the highest recall at **76.96%**, the highest F1-score at **0.1022**, and the strongest discriminative capability with an **ROC-AUC of 0.8495**.

Logistic Regression was selected as the final operational model. In insurance risk management, the business cost of a False Negative (failing to identify a genuine claim) far exceeds that of a False Positive (flagging a non-claim for review). Logistic Regression successfully captured nearly three out of four claims, making it the most viable model for deployment.

Complementing the supervised pipeline, **K-Means clustering** was implemented as an unsupervised exploratory technique. Using the Elbow Method, **five clusters (\$k=5\$)** were identified, and Principal Component Analysis (**PCA**) was applied for two-dimensional visual projection. This clustering analysis revealed distinct groupings across customer demographics, travel agencies, and policy structures, confirming the presence of latent behavioral profiles within the insured population.